AI-ENABLED WAREHOUSE DOCUMENT MANAGEMENT SYSTEM

Conceptual Framework

The conceptual framework (Figure 1) illustrates the integration of **Artificial Intelligence (AI), secure document processing, role-based access control, and workflow automation** to enhance efficiency, accuracy, and transparency in warehouse document management.

The framework demonstrates how documents flow through the system from submission to archiving while being processed by AI components for **classification, validation, analysis, and decision support**. Unlike traditional systems, the proposed model introduces intelligent automation to reduce manual workload and improve operational performance.

Each warehouse must access its specific documents, while regulatory body

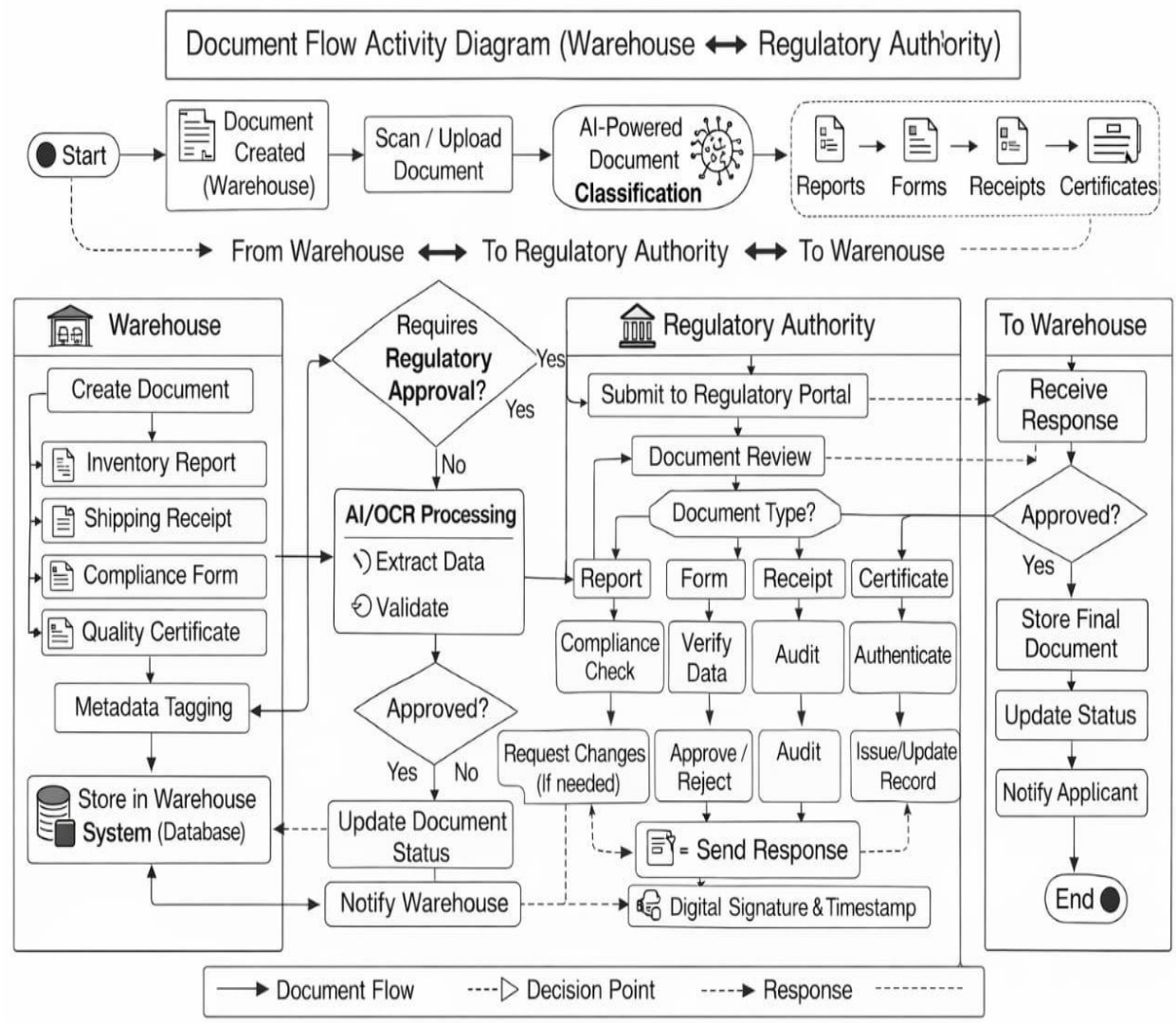


Figure 1: Activity diagram flow of whole system including AI

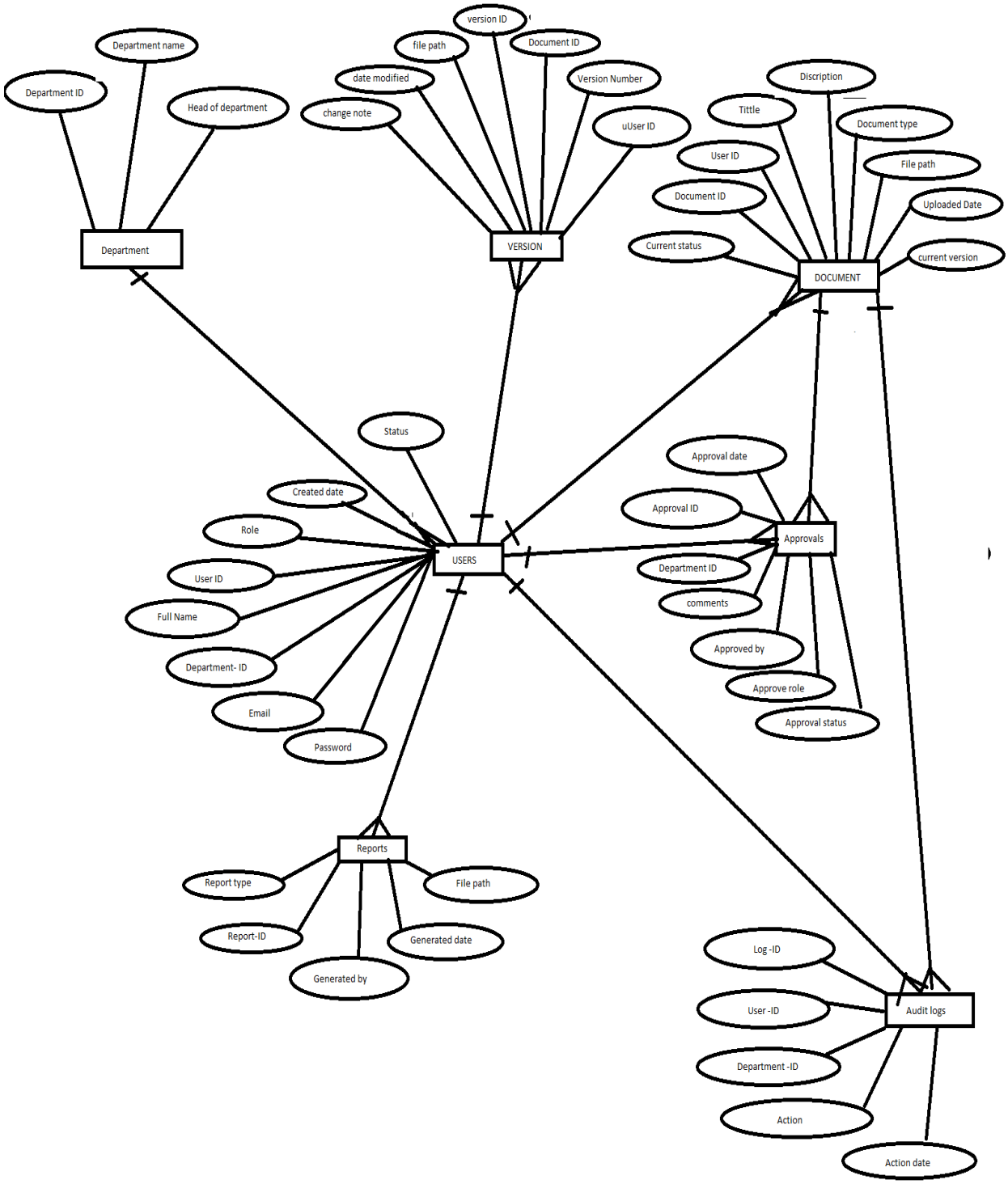


Figure 2: Diagram of ERD

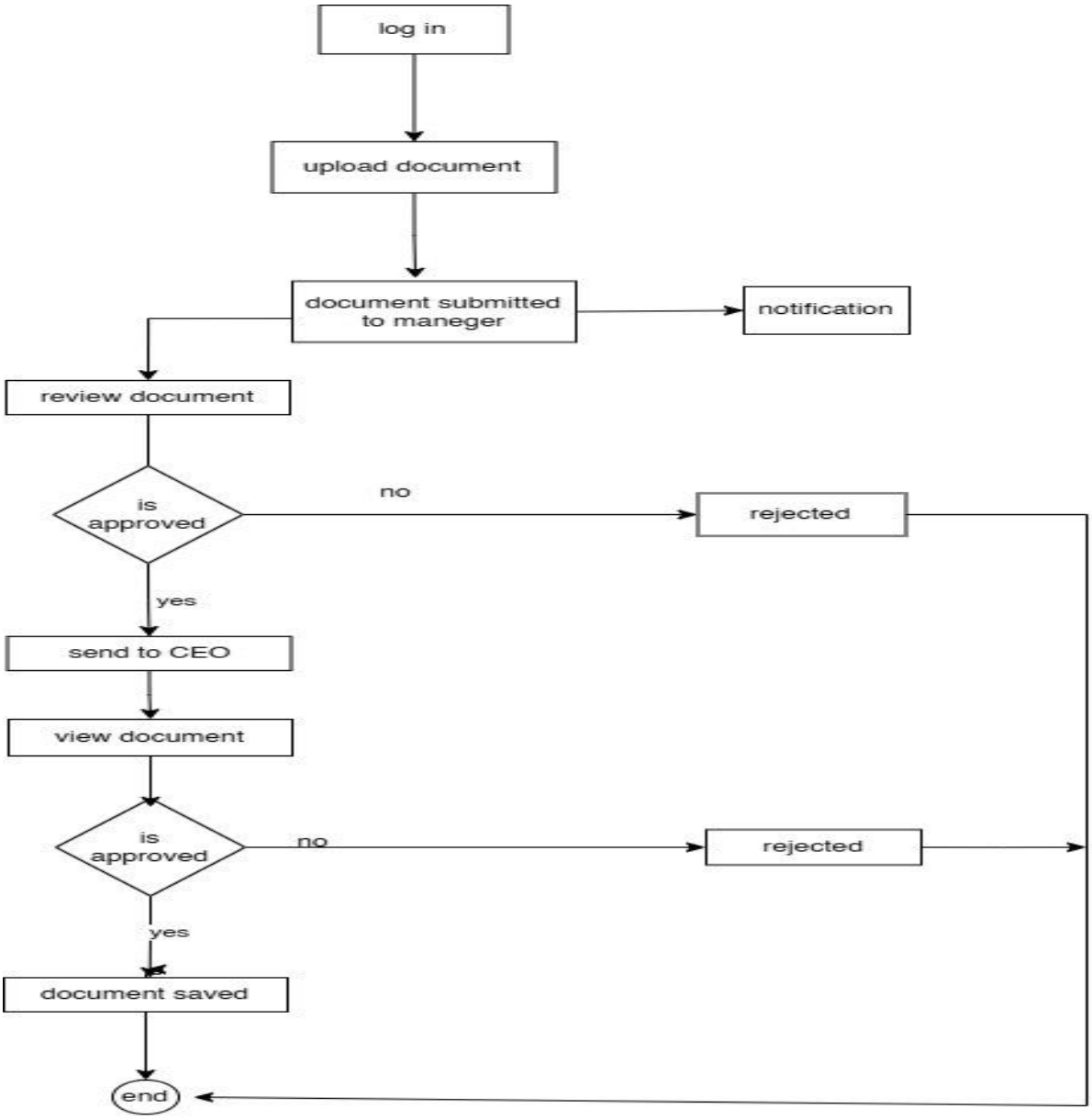


Figure 3:Activity diagram for internal flow of documents

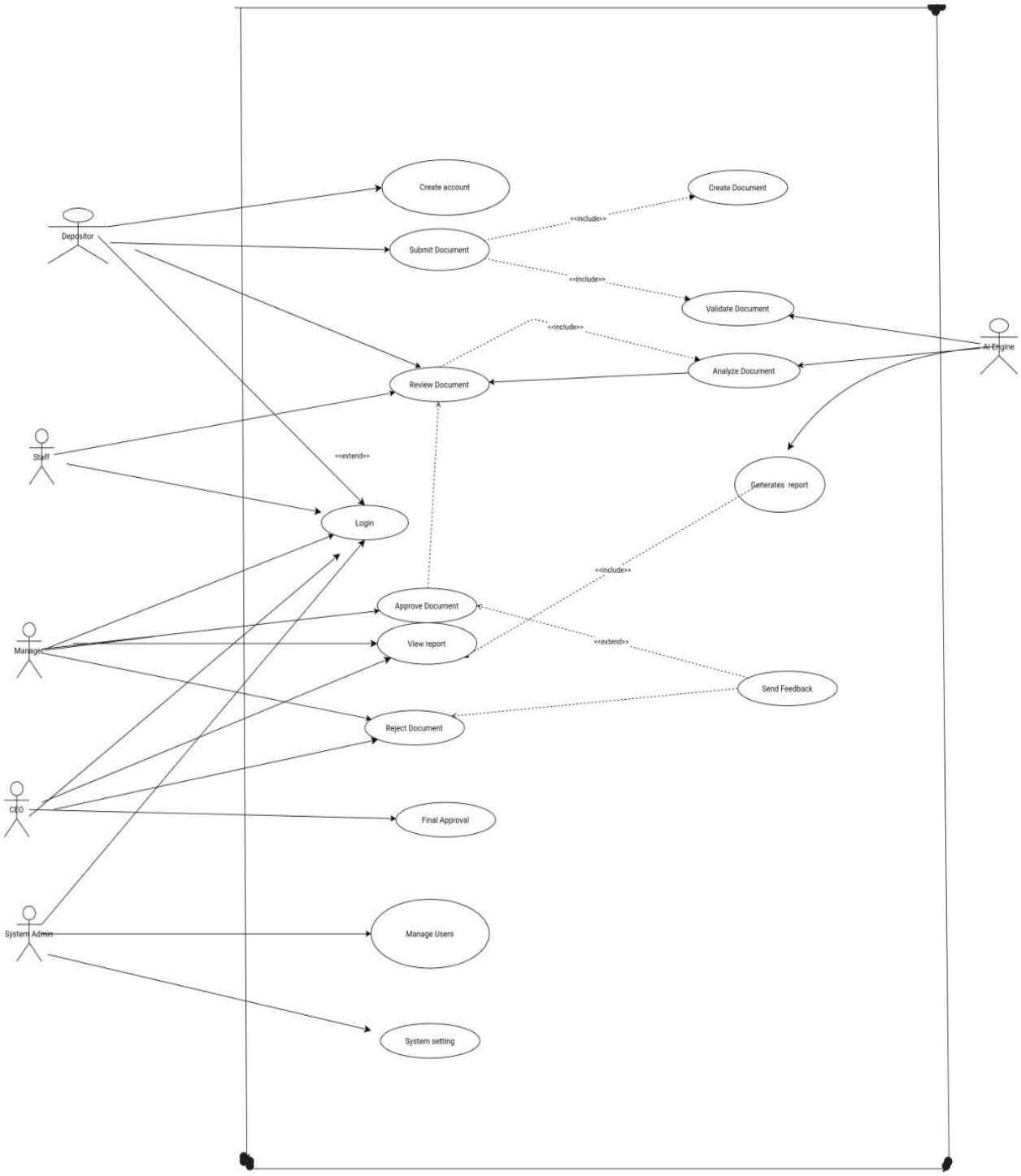


Figure 4: USE CASE diagram

Key Components

• Document Submission

Warehouse users submit documents through a secure web interface. The system captures document metadata such as title, category, uploader identity, and timestamp for tracking and indexing.

• **AI Processing Engine**
The system integrates AI technologies to automate document handling:

- **OCR (Optical Character Recognition):** Extracts text from scanned documents
- **Machine Learning Classification:** Categorizes documents automatically
- **Natural Language Processing (NLP):** Enables semantic understanding and intelligent search
- **Semantic search or preview**
- **Generation and analysis of report**
- **Validation and Error Detection:** Ensures document completeness and correctness
- **Recommendation Engine:** Suggests approval or rejection decisions

• **Approval Workflow System**
A multi-level approval mechanism ensures accountability:

- **Staff:** Performs initial verification
- **Manager:** Reviews and makes approval decisions
- **CEO:** Conducts final validation and approval

• **Reporting and Analytics**
The system automatically generates analytical reports, enabling data-driven decision-making and performance monitoring.

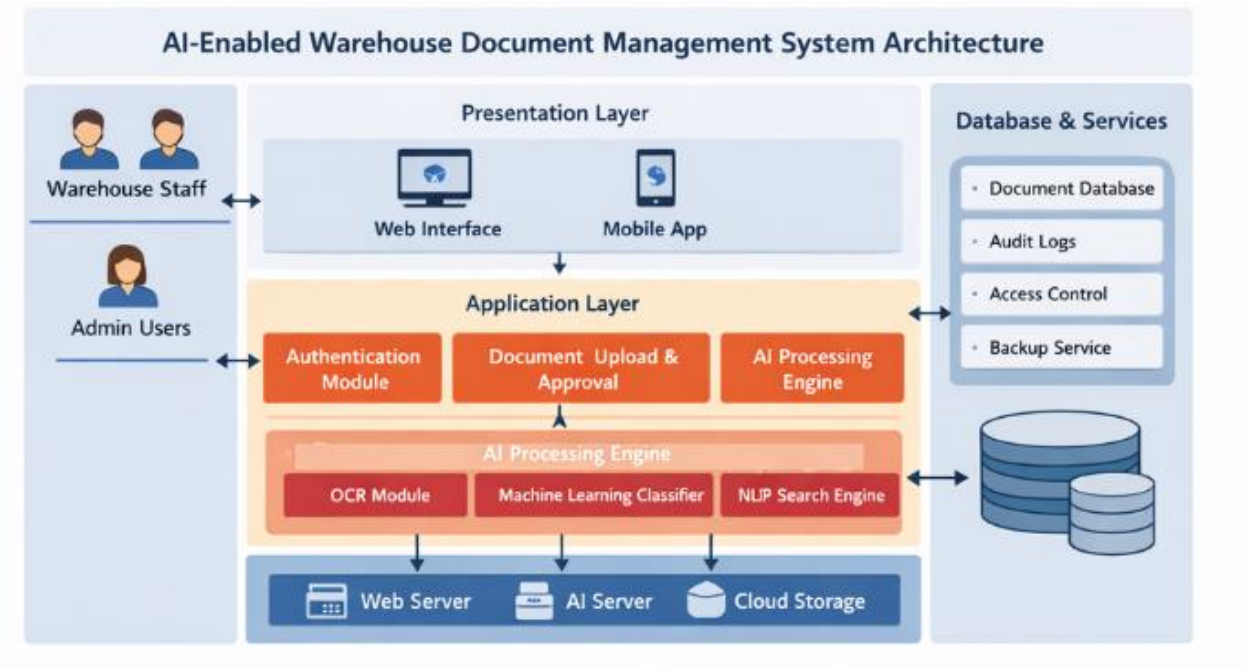
• **Access Control System**
Role-based access control (RBAC) ensures that users only access functionalities relevant to their roles, enhancing system security and integrity.

• **Notification and Feedback System**
Automated notifications inform users about document status changes, approvals, rejections, and required modifications such as email or text

NOTE: When the depositor performs a search in the system, they should be provided with an option to select nearby (location) warehouses. The system should display a list of available warehouses, allowing the depositor to choose one from the list**.**

System Overview and Architecture

The proposed system adopts a **three-tier architecture** consisting of the Frontend Layer, Backend Layer, and AI Processing Layer to ensure modularity, scalability, and efficient system operation.



Frontend Layer (User Interface)

The frontend layer provides an interactive interface for different user roles:

Depositor – Submits warehouse documents

Staff – Registers and processes documents

Manager – Reviews and approves documents

CEO – Final approval and reporting

NOTE: Every user should be able to create, upload, submit documents

Documents within the warehouse follow a structured internal workflow to ensure proper verification and approval. A document is first created and submitted by the depositor, then reviewed by staff for initial verification. After that, it is forwarded to the manager for decision-making and finally to the CEO for final approval. At each stage, feedback can be provided and sent back to the previous level if corrections are required, ensuring accuracy and completeness before finalization.

Once the document is fully approved, it establishes a link between the warehouse and regulatory authorities. Approved documents such as inspection reports, compliance certificates, and

transaction records are shared with regulators for monitoring, auditing, and compliance purposes. This connection ensures transparency, accountability, and adherence to regulations, allowing regulatory bodies to verify warehouse operations based on authenticated and approved documents

System Administrator – Manages system users and configuration

Phase 1 — System Access (Login Process)

Step 1: User Login

All users must log into the system before performing any task.

Actors involved:

Depositor, Staff, Manager, CEO, System Administrator

Process

User opens system login page

User enters: Username, Password

System verifies credentials, System identifies user role, System loads role-based dashboard

Output

Depositor Dashboard, Staff Dashboard, Manager Dashboard, CEO Dashboard, Admin Dashboard

Phase 2 — Document Submission (Depositor)

Actor: Depositor

Step 2: Depositor Uploads Document

Depositor submits warehouse documents into the system.

Process

Depositor logs into system

Depositor opens Submit Document page

Depositor fills form:

Document title

Document type

Description

Date

Upload file (PDF/Image)

Depositor clicks Submit

System Actions

System automatically:

Saves document into database

Stores file on server

Assigns Document ID

Sets status:

Status = Pending Staff Review

Records submission in audit log

Output

Documents should remain in depositor dashboard and if needed should be allowed to download

Document appears in:

Staff Dashboard → Pending Documents

Phase 3 — Document Registration (Staff)

Actor: Staff

Step 3: Staff Receives Document

Staff processes documents submitted by depositors.

Process

Staff logs into system

Staff opens:

Pending Documents

Staff selects document

Staff reviews document details

Staff Actions

Staff may:

Edit document information

Verify document

Add notes

Confirm document

System AI Processing

After staff confirmation:

System automatically:

Runs OCR

Extract text from document

Runs Classification Model

Assign category automatically

Examples:

Import document, Export document, Storage document, generates keywords, Saves metadata Status Update, System changes status**: Pending Manager Approval**

Output

Document appears in:

Manager Dashboard

Phase 4 — Document Approval (Manager)

Actor: Manager

Step 4: Manager Reviews Document

Manager evaluates documents before final approval.

Process

Manager logs into system

Manager opens:

Documents Awaiting Approval

Manager selects document

Manager Actions

Manager may:

✓Approve Document, ✓ Reject Document, ✓ Request Correction

If Approved

System updates status: Pending CEO Approval

If Rejected

System updates status: Rejected

Depositor receives notification.

Output

Approved documents appear in: CEO Dashboard

Phase 5 — Final Authorization (CEO)

Actor: CEO

Step 5: CEO Final Approval

CEO performs final authorization.

Process

CEO logs into system

CEO opens: Documents Awaiting Final Approval

CEO reviews document

CEO Actions

CEO may: ✓ Final Approve, ✓ Reject

If Approved

System updates: Status = Approved

Document becomes official warehouse record.

System Actions

System automatically: Archives document, enables search, Adds audit record. Generates report entry

Output

Document appears in: Approved Documents

Reports

Search System

Phase 6 — Document Retrieval (All Users)

Actors

Staff, Manager, CEO

Step 6: Search Document

Users retrieve stored documents.

Process

User enters: Document number,

Keyword

Category

System AI Actions

System uses:

✓ NLP search✓ Keyword search

Output

Matching documents displayed.

User can: View, Download, Print

Phase 7 — Reporting

Actors: Manager, CEO

Step 7: Generate Reports

Process

User selects:

Date range

Document type

Status

Output

System generates:

Document reports

Approval reports

Activity reports

Phase 8 — System Administration

Actor: System Administrator

Step 8: Manage System

Administrator controls system configuration.

Admin Functions

Admin can:✓ Create users

✓ Edit users

✓ Delete users

✓ Assign roles

Roles: Staff

Depositor

Manager

CEO

Admin Also Manages

System settings

Backup

Security

Complete Workflow Sequence

Step-by-Step Order

Admin creates users

User logs in

Depositor uploads document

Staff verifies document

System runs AI processing

Manager approves document

CEO final approval

Document archived

Users search document

Reports generated

Simple Developer Logic Flow

LOGIN

↓

ROLE IDENTIFICATION

↓

DEPOSITOR UPLOAD

↓

STAFF VERIFICATION

↓

AI PROCESSING

↓

MANAGER APPROVAL

↓

CEO APPROVAL

↓

DOCUMENT ARCHIVE

↓

SEARCH & REPORTING

TYPSE OF DOCUMENTS

FORMS

Application form(depositor>staff)

Permission form(staff>manager>CEO)

Warehouse inspector(compliance) form (staff should have a physical form and after performing inspection of a certain warehouse then will be uploaded to the system then every warehouse

should be able to download and system should generate a general report for all warehouse that will be used for regulatory ranking by AI)

RECEIPTS

Warehouse receipt delivery (staff >depositor)

Goods cost receipt notes(staff>depositor)

CERTIFICATES

Quality certificates for depositor(warehouse>depositor)

Warehouse compliance certificates (regulatory body>warehouse)

REPORTS

Application report (staff>manager>CEO)

Ranking report (regulatory body>warehouse)

Inspection report (regulatory body>warehouse)

Warehouse operation cost structure reports(warehouse>regulatory)